

Results outline

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Lay abstract

Shifted phenology with climate change has extended the vegetative growing season, which is widely expected to increase tree growth. Multiple recent studies, however, have failed to find this relationship and suggested shifts in drought or competition may prevent increased growth. Here, we ask if trees grow more with longer seasons in an environment where drought and competition are limited. We aim to answer this question by leveraging 9 years of vegetative phenology data from two studies located in an urban arboretum and by relating it to tree growth (Figure 1). We found contrasting evidence that trees grow more during longer seasons. Nine of our species did not change their growth; one grew less at its adult stage but more at its juvenile stage, and two additional species grew more with longer seasons. These responses increased when using the calendar instead of the thermal growing season. Our main findings demonstrate that tree growth responses with climate change may be more complex than previously suggested. The shifts in response to season lengths were not massive, and when they happened it was from juvenile more than mature trees.

Research questions

1. Did growth change across years, and if so what cue had the largest effects?
2. How do species ring widths vary with thermal season vs calendar season?
3. Does the previous year cumulated GDD influence the following year's growth?
4. How do species spring and autumn phenophases shift with the temperature of different seasons?
5. Does growth change across sites?

Results

1. Summary results
 - (a) Wildchokkie includes four deciduous tree species from four sites with 227 complete growing seasons (GS)
 - (b) Phenology data span three years from 2018 to 2020 with 239 leafout and 230 budset observations
 - (c) The earliest leafout day of year was 112 (2019) and the latest, 157 (2019).
 - (d) The earliest budset day of year was 225 (2018) and the latest, 297 (2018)
 - (e) GS thermal sum ranged from 1465.1 GDD (*B. alleghaniensis* in 2018) to 2513.6 GDD, (*A. incana* in 2018)
 - (f) GSL ranged from 90 days (*B. alleghaniensis* in 2018) to 173 days (*A. incana* in 2018)
 - (g) Annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018).
2. Did growth change across years, and if so what cue had the largest effects?
 - (a) Growth was highest on average in 2018, but no cue dominated (it was neither: the wettest; the driest; the shortest; the latest SOS; the earliest EOS or highest thermal sum)
 - (b) Growth was lowest on average in 2020 and that year was: the driest; the shortest; the latest SOS and EOS and had the highest thermal sum

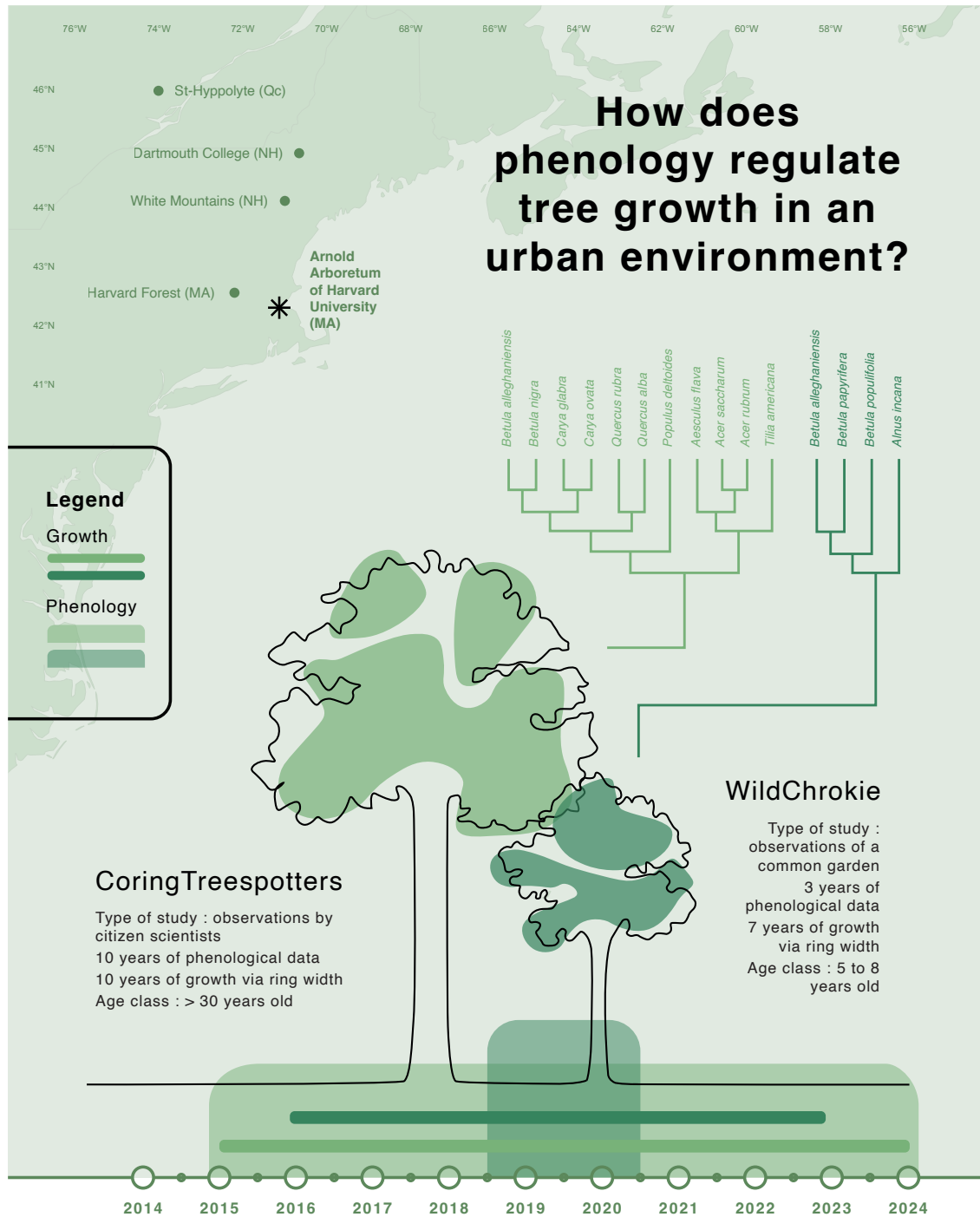


Figure 1: Overview of the age class, species, provenance of the trees used in each study along with the type of study each project consists of. The colored lines represent the period of the growth data and the shaded areas are the phenological data. Both are colour-coded by project. Each phylogram illustrates the species used in each project. The map illustrates the four different provenances used in Wildchrokrie, and the star represents the location of Wildchrokrie and CoringTreespotters.

3. How do species ring widths vary with thermal season vs calendar season?

- A later start of season decreased growth for two species (*A. incana*: $\beta = -0.13$, $UI_{90} [-0.16, -0.09]$ and *B. populifolia*: $\beta = -0.09$, $UI_{90} [-0.16, -0.02]$) (Figure 2c (Table S1)).
- A later end of season decreased growth only for *A. incana* ($\beta = -0.07$, $UI_{90} [-0.11, -0.03]$) (Figure 2d (Table S1)).

- (c) Longer thermal season increased growth only for *B. papyrifera* ($\beta = 0.32$, UI_{90} [0.04, 0.6]) (Figure 2a (Table S1).
 - (d) But longer calendar season increased growth for two additional species (*A. incana*: $\beta = 0.13$, UI_{90} [0.05, 0.21], *B. alleghaniensis*: $\beta = 0.1$, UI_{90} [0, 0.19] and *B. papyrifera*: $\beta = 0.26$, UI_{90} [0.06, 0.46]) (Figure 2b (Table S1).
 - (e) Calendar season had a bigger effect on growth than the thermal season (Figure S1)
4. Does the previous year cumulated GDD influence the following year's growth?
 - (a) Growth decreased for all species with current year GDD
 - (b) But growth decreased for only two species with previous year GDD
 5. How do species spring and autumn phenophases shift with the temperature of different seasons?
 - (a) *results in progress so this is just an example of where it would belong*: warmer winter delayed leafout, but warmer springs advanced leafout.
 6. Does growth change across sites?
 - (a) Growth did not change across any site (Figure 3) (Table S2).

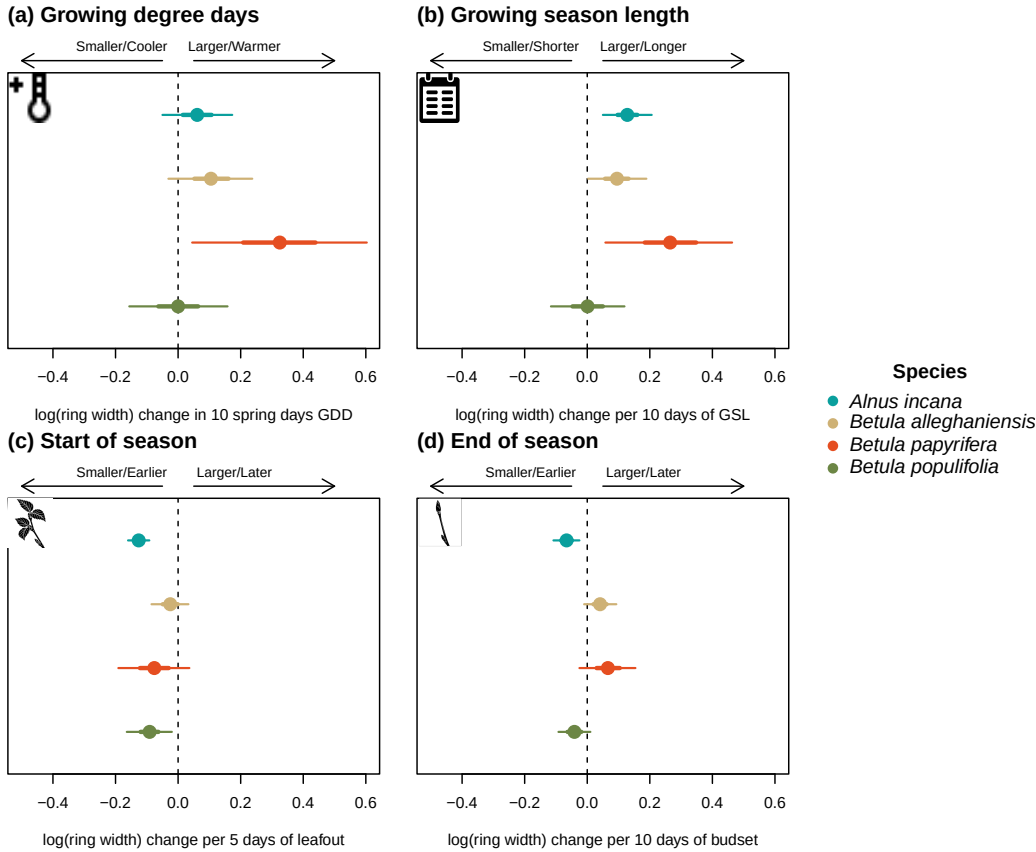


Figure 2: Model slope parameters across the different model predictors for Wildchrokie species. GDD represent the accumulated growing degree days between leafout and budset (a) the models estimates shows log ring width change per 10 average spring days GDD (174.32). GSL is the number of days between leafout and budset (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through budset (d). All predictors were fit using the same model and number of observations, with only the independent variable changing. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

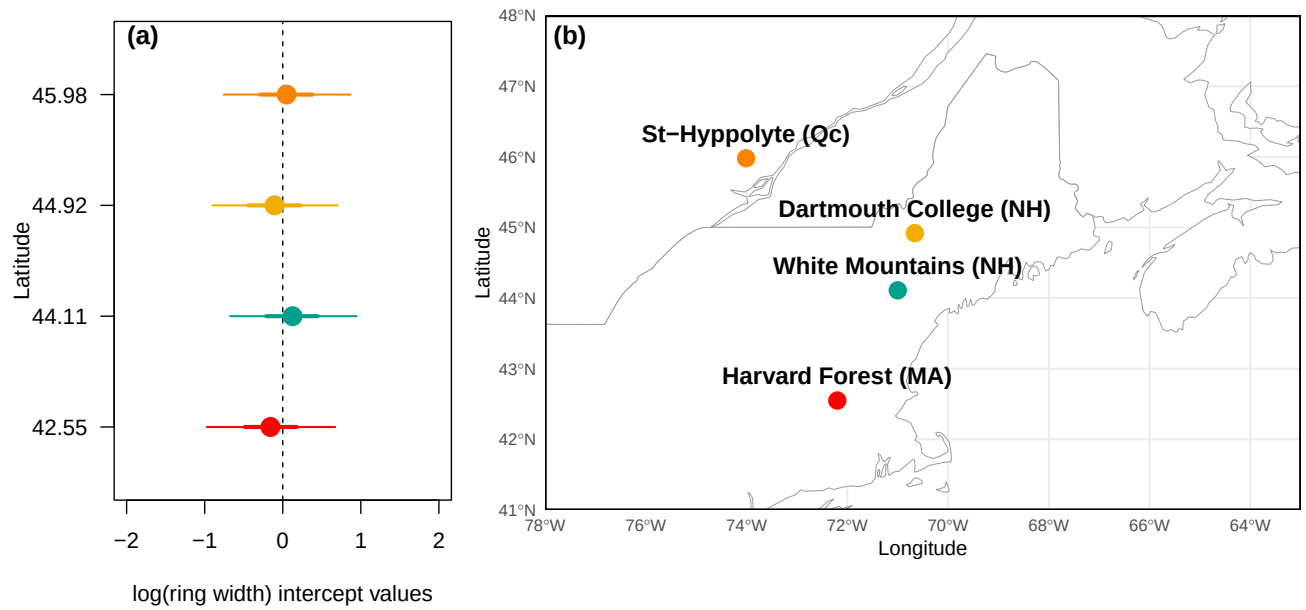


Figure 3: The effect of sites on growth across a latitudinal gradient. (a) Model intercept parameters for each site with growing degree days (GDD) as the predictor. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI. The sites are color-coded from (b), the map showing their locations.

Supplemental results

Table S1: Species level slope parameters (β and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Species	GDD	GSL	SOS	EOS
<i>A. incana</i>	$\beta = 0.06, [-0.05, 0.17]$	$\beta = 0.13, [0.05, 0.21]$	$\beta = -0.13, [-0.16, -0.09]$	$\beta = -0.07, [-0.11, -0.03]$
<i>B. alleghaniensis</i>	$\beta = 0.10, [-0.03, 0.24]$	$\beta = 0.10, [0.00, 0.19]$	$\beta = -0.03, [-0.09, 0.03]$	$\beta = 0.04, [-0.01, 0.09]$
<i>B. papyrifera</i>	$\beta = 0.33, [0.04, 0.60]$	$\beta = 0.27, [0.06, 0.46]$	$\beta = -0.08, [-0.19, 0.04]$	$\beta = 0.07, [-0.03, 0.15]$
<i>B. populifolia</i>	$\beta = 0.00, [-0.16, 0.16]$	$\beta = 0.00, [-0.12, 0.12]$	$\beta = -0.09, [-0.16, -0.02]$	$\beta = -0.04, [-0.09, 0.01]$

Table S2: Site level intercept parameters (α and 90% uncertainty intervals, in square brackets) for the GDD model.

Site	Intercept.estimate	Latitude	Longitude
Dartmouth College (NH)	$\alpha = -0.11, [-0.9, 0.7]$	44.79	-71.15
Harvard Forest (MA)	$\alpha = -0.16, [-0.97, 0.67]$	42.53	-72.19
St-Hyppolyte (Qc)	$\alpha = 0.05, [-0.76, 0.87]$	45.93	-74.03
White Mountains (NH)	$\alpha = 0.13, [-0.67, 0.95]$	44.11	-71.23

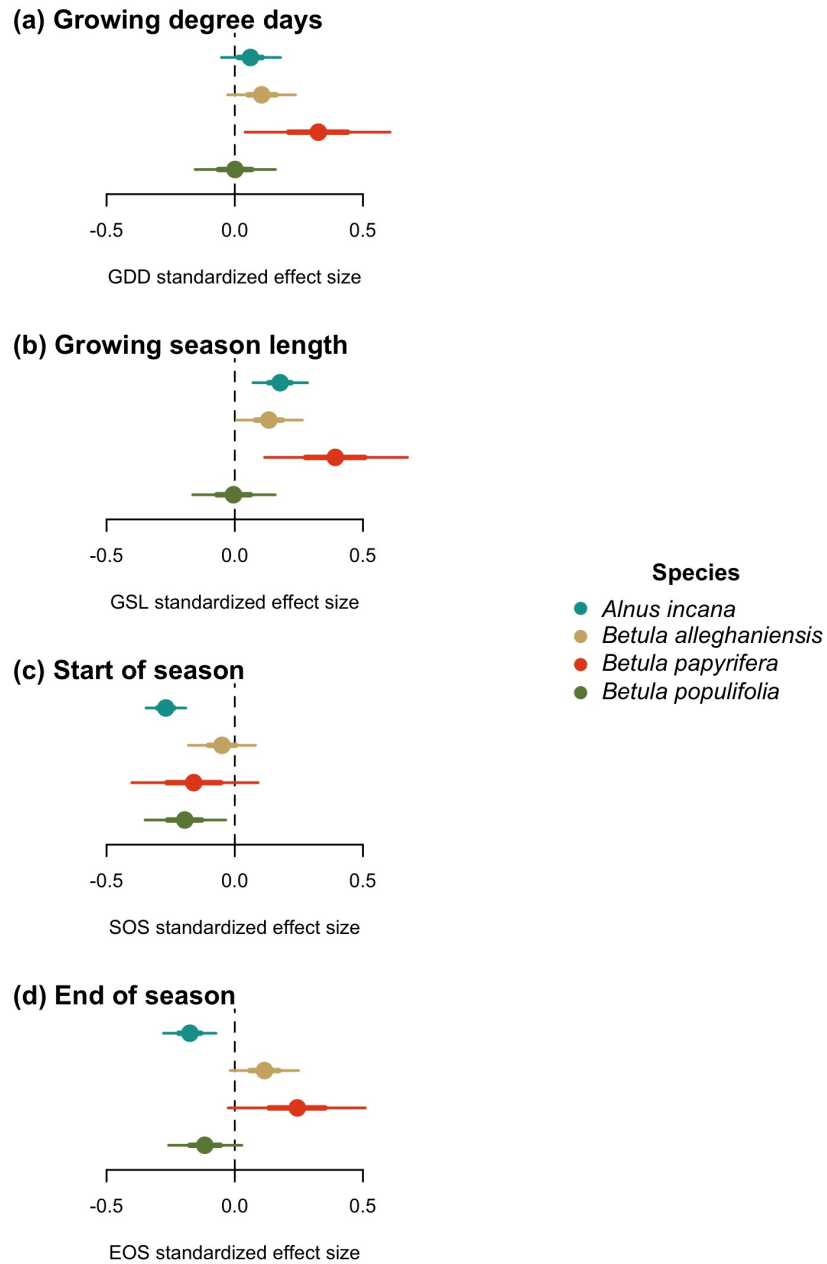


Figure S1: Standardized (z-scored) slope estimates for Wildchrokie. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.